R91/B_utility_probabilities | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 28 | update 40320 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	71.0914	66.2037	11.416997	11.427978	11.371692	-0.1909	+0.3968	0.4084
TEST ALL	71.4086	66.1802	11.374209	11.385299	11.329932	-0.1942	+0.3890	0.3990

Dataset	n	NSS Top1 (%)	R91 Top1 (%)	NSS cost	R91 cost
TSP TEST	4500	45.888889	47.333333	7.970400719	7.971633684
CVRP TEST	4500	45.977778	48.044444	18.004802368	18.015935162
TSPLIB	49	20.408163	30.612245	8.217907074	8.192447597
CVRPLIB	100	70.000000	62.000000	67.101111717	67.279303484

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS TEST predictions were freshly computed on these4500 instances; old3895 TEST metrics are not reused.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: original_score_mlp_maximin | objective: legal-masked utility CE

R91/B_utility_probabilities | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.0914	66.2037	11.416997	11.427978	11.371692	-0.1909	+0.3968	0.4084	per-problem TRAIN SBS
TSP	46.9778	33.1111	8.025374	8.045354	7.990533	-0.2483	+0.4360	0.4253	T2T
CVRP	47.1333	18.7111	18.165104	18.269377	18.064087	-0.5708	+0.5592	0.7510	RELD_CVRP
ATSP	76.9111	47.4889	1.638938	1.676905	1.632471	-2.2641	+0.3962	0.3938	UNICO_MatPOENet
OVRP	40.9333	35.7556	7.461986	7.469579	7.415777	-0.1017	+0.6231	0.6359	MoSES_RF
VRPB	81.2222	80.8444	9.353010	9.354876	9.318750	-0.0200	+0.3676	0.3759	reld_mt
VRPL	37.8889	33.0889	12.432164	12.437959	12.383444	-0.0466	+0.3934	0.4008	MoSES_RF
VRPTW	80.0000	79.9556	18.818942	18.819431	18.766139	-0.0026	+0.2814	0.2975	reld_mt
OVRPTW	83.8444	83.7778	10.143061	10.143730	10.116859	-0.0066	+0.2590	0.2563	reld_mt
OVRPB	88.4222	88.5111	6.357921	6.357254	6.341078	+0.0105	+0.2656	0.2786	reld_mt
OVRPL	41.1778	35.9111	7.438917	7.445709	7.390416	-0.0912	+0.6563	0.6734	MoSES_RF
VRPBL	78.8444	78.5111	9.367640	9.371523	9.328366	-0.0414	+0.4210	0.4241	reld_mt
VRPBTW	78.5333	78.4222	19.947418	19.950430	19.851084	-0.0151	+0.4853	0.4652	reld_mt
VRPLTW	78.7556	78.7556	18.763665	18.765871	18.707843	-0.0118	+0.2984	0.3150	reld_mt
OVRPBL	89.3333	89.2000	6.368933	6.370093	6.352824	-0.0182	+0.2536	0.2634	reld_mt
OVRPBTW	83.4889	83.4444	10.661994	10.661098	10.623245	+0.0084	+0.3648	0.3422	reld_mt
OVRPLTW	83.4889	83.4889	10.157080	10.157423	10.129818	-0.0034	+0.2691	0.2747	reld_mt
VRPBLTW	79.4889	79.4000	19.793223	19.797044	19.706420	-0.0193	+0.4405	0.4284	reld_mt
OVRPBLTW	83.2000	83.2889	10.610571	10.609955	10.571302	+0.0058	+0.3715	0.3494	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R91/B_utility_probabilities | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.4086	66.1802	11.374209	11.385299	11.329932	-0.1942	+0.3890	0.3990	per-problem TRAIN SBS
TSP	47.3333	34.5333	7.971634	7.989621	7.937593	-0.2251	+0.4288	0.4222	T2T
CVRP	48.0444	17.9333	18.015935	18.124983	17.918509	-0.6016	+0.5437	0.7253	RELD_CVRP
ATSP	76.6000	47.2444	1.635722	1.674207	1.629441	-2.2987	+0.3854	0.3856	UNICO_MatPOENet
OVRP	40.9111	33.6444	7.445601	7.455750	7.400487	-0.1361	+0.6096	0.6201	MoSES_RF
VRPB	80.6000	80.4000	9.349957	9.351810	9.315298	-0.0198	+0.3721	0.3751	reld_mt
VRPL	38.2222	32.2667	12.349965	12.356138	12.300958	-0.0500	+0.3984	0.4065	MoSES_RF
VRPTW	81.2667	80.8444	18.687998	18.690545	18.637408	-0.0136	+0.2714	0.2816	reld_mt
OVRPTW	83.9556	83.9333	10.092671	10.093063	10.068135	-0.0039	+0.2437	0.2345	reld_mt
OVRPB	88.0667	88.1333	6.370728	6.370826	6.352225	-0.0015	+0.2913	0.2977	reld_mt
OVRPL	42.2444	34.6000	7.463790	7.472542	7.418033	-0.1171	+0.6168	0.6278	MoSES_RF
VRPBL	79.6444	79.4222	9.395102	9.396379	9.355967	-0.0136	+0.4183	0.4183	reld_mt
VRPBTW	78.8222	78.7556	19.665261	19.665725	19.577666	-0.0024	+0.4474	0.4438	reld_mt
VRPLTW	79.2444	79.2444	18.803373	18.803136	18.745143	+0.0013	+0.3106	0.3294	reld_mt
OVRPBL	89.2667	89.2889	6.375766	6.375741	6.359310	+0.0004	+0.2588	0.2701	reld_mt
OVRPBTW	84.8889	84.8222	10.660422	10.659591	10.624748	+0.0078	+0.3358	0.3121	reld_mt
OVRPLTW	82.9556	82.8667	10.085609	10.086112	10.059001	-0.0050	+0.2645	0.2659	reld_mt
VRPBLTW	79.7111	79.7333	19.704403	19.707068	19.613284	-0.0135	+0.4646	0.4477	reld_mt
OVRPBLTW	83.5778	83.5778	10.661820	10.662149	10.625574	-0.0031	+0.3411	0.3178	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R91/B_utility_probabilities | LIB and locked provenance

Current R91 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	30.6122	10.2041	8.192448	8.240435	8.142167	-0.5823	+0.6175	0.6029
CVRPLIB	62.0000	45.0000	67.279303	67.245364	66.945093	+0.0505	+0.4992	0.6036

SHA256: cdcb5f1da1e3c30c59bb751ad0f8a019416e8b56d41ab36928dd2ee9de86296d

Unique selection: strict minimum full VAL macro_vs_sbs_pct with common new81000 TRAIN-fitted SBS; first wins ties

Inference: eval mode, fixed physical-node0 TSP slot3 role, original R77 score MLP/pair differences/maximin; stable ascending-global-solver-ID tie break

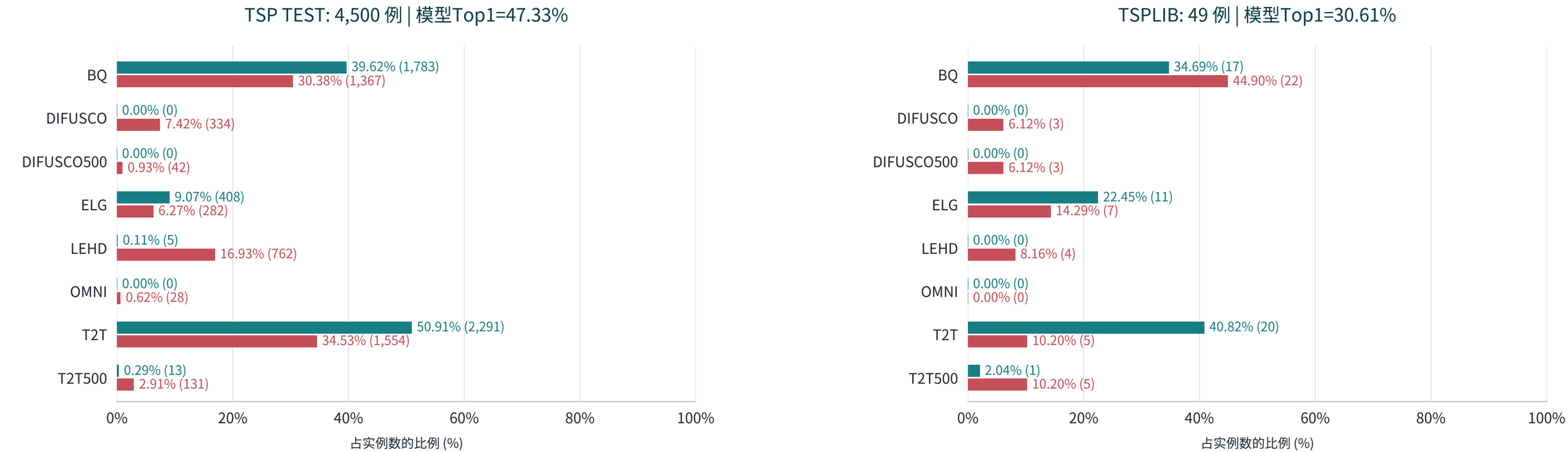
Normalization: new81000 TRAIN only; no TEST/LIB refit.

R91/B_utility_probabilities | TSP | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 cdcb5f1da1e3c30c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

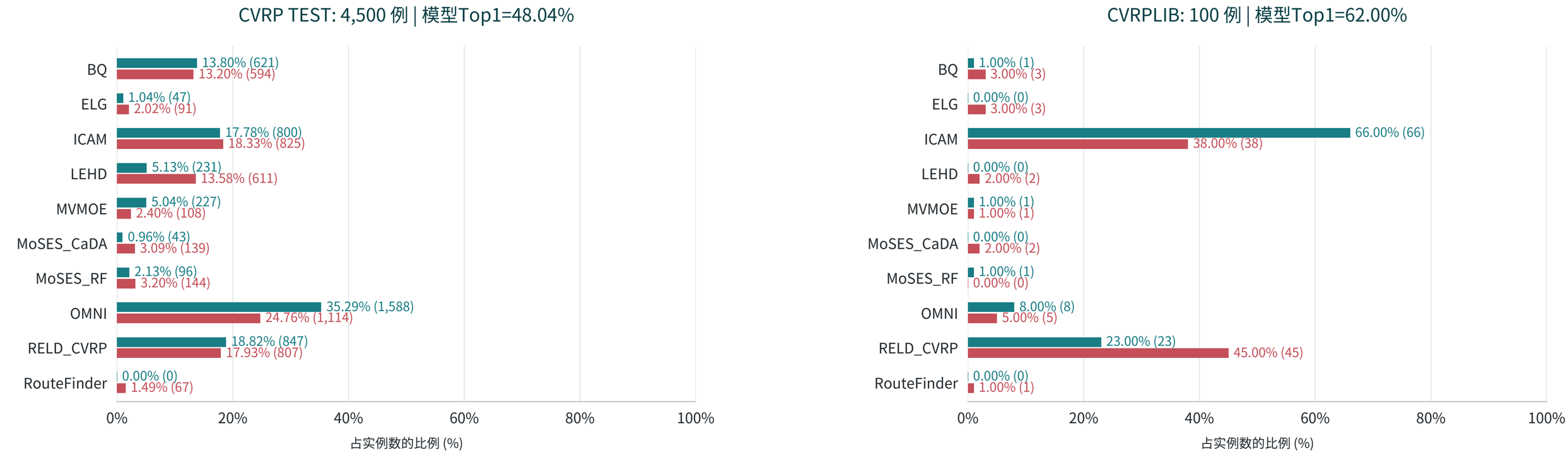
R91/B_utility_probabilities: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/B_utility_probabilities | CVRP | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 cdcb5f1da1e3c30c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

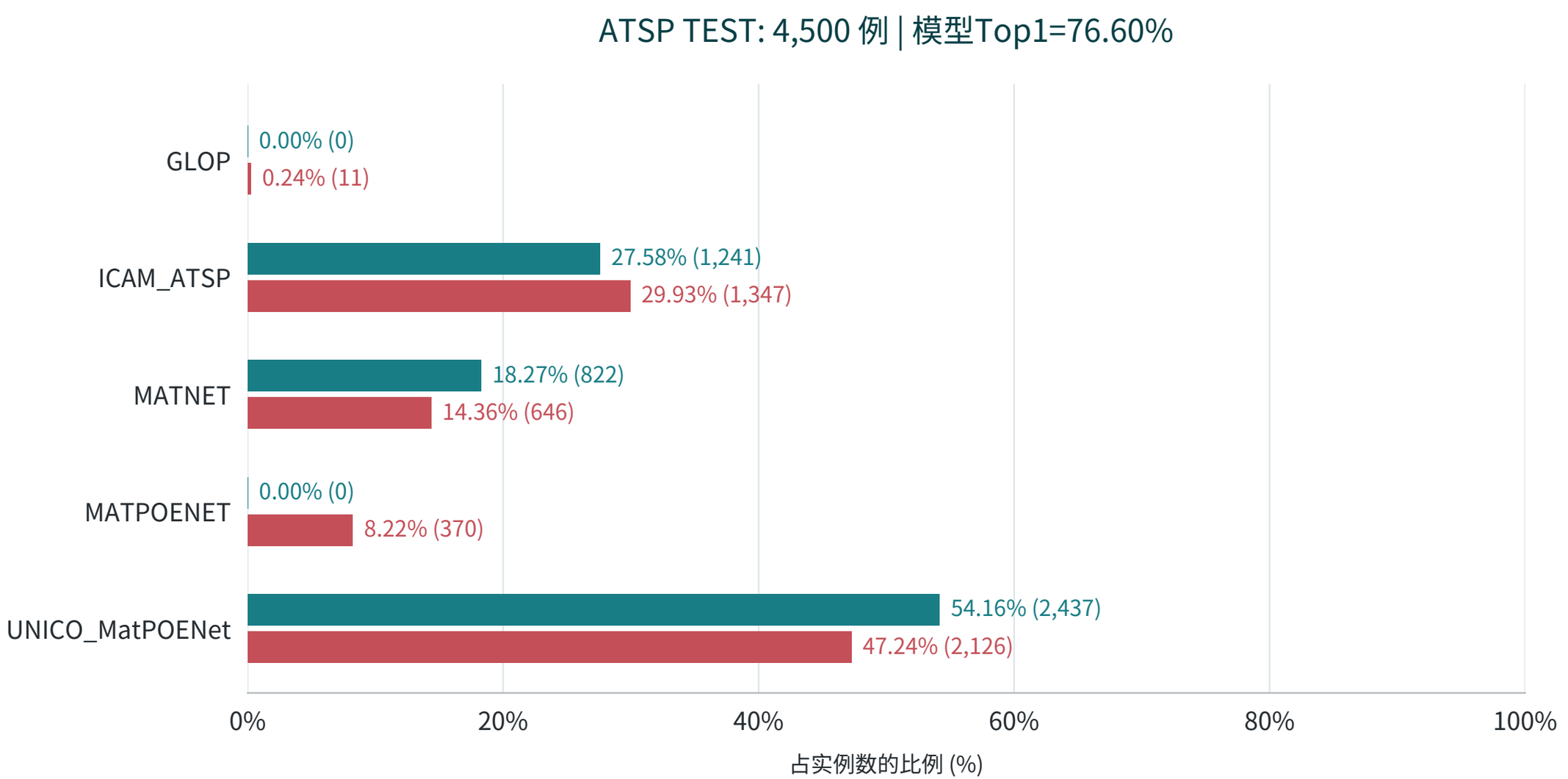
R91/B_utility_probabilities: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/B_utility_probabilities | ATSP | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 cdcb5f1da1e3c30c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

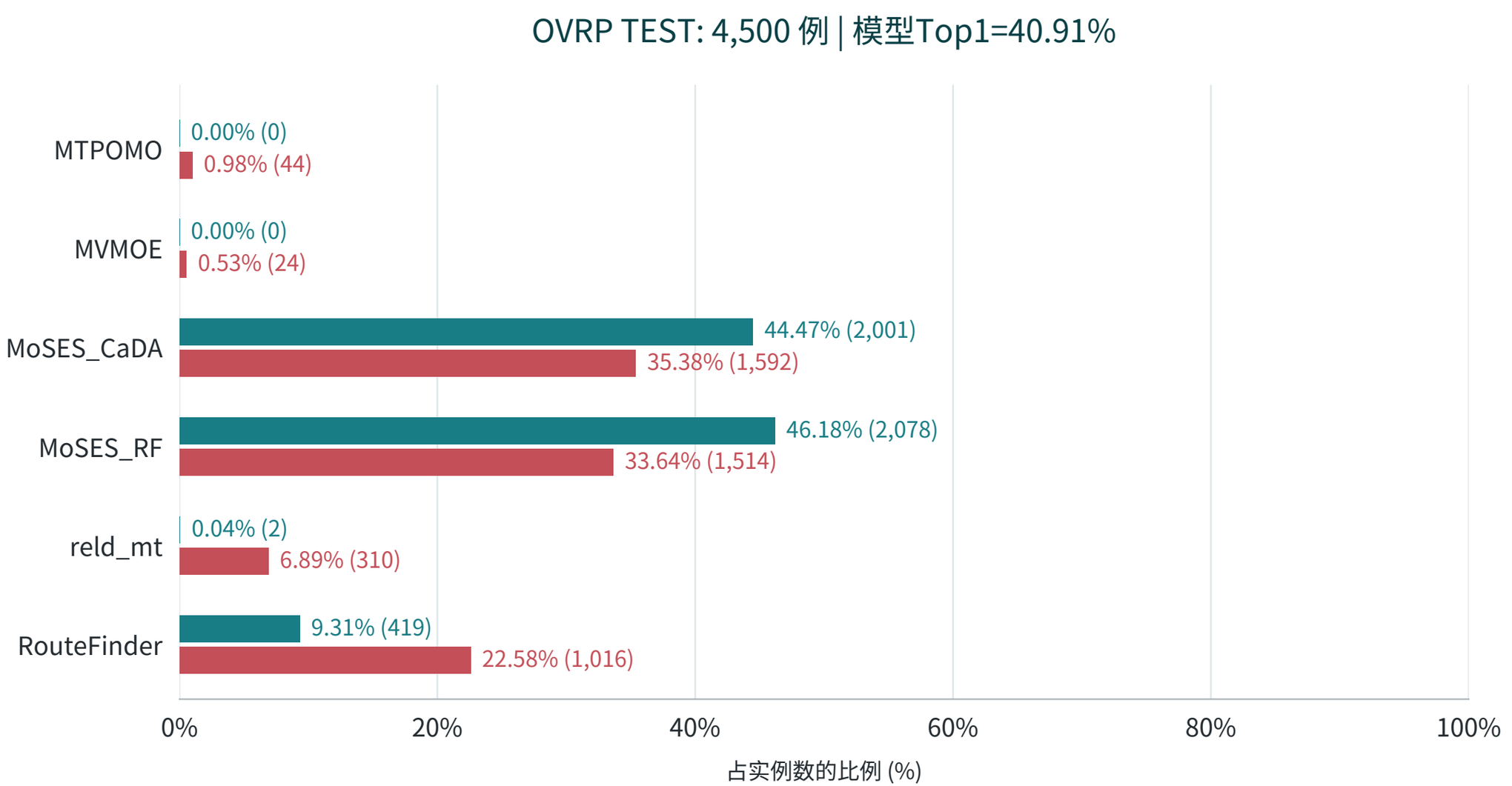
R91/B_utility_probabilities: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/B_utility_probabilities | OVRP | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 cdcb5f1da1e3c30c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

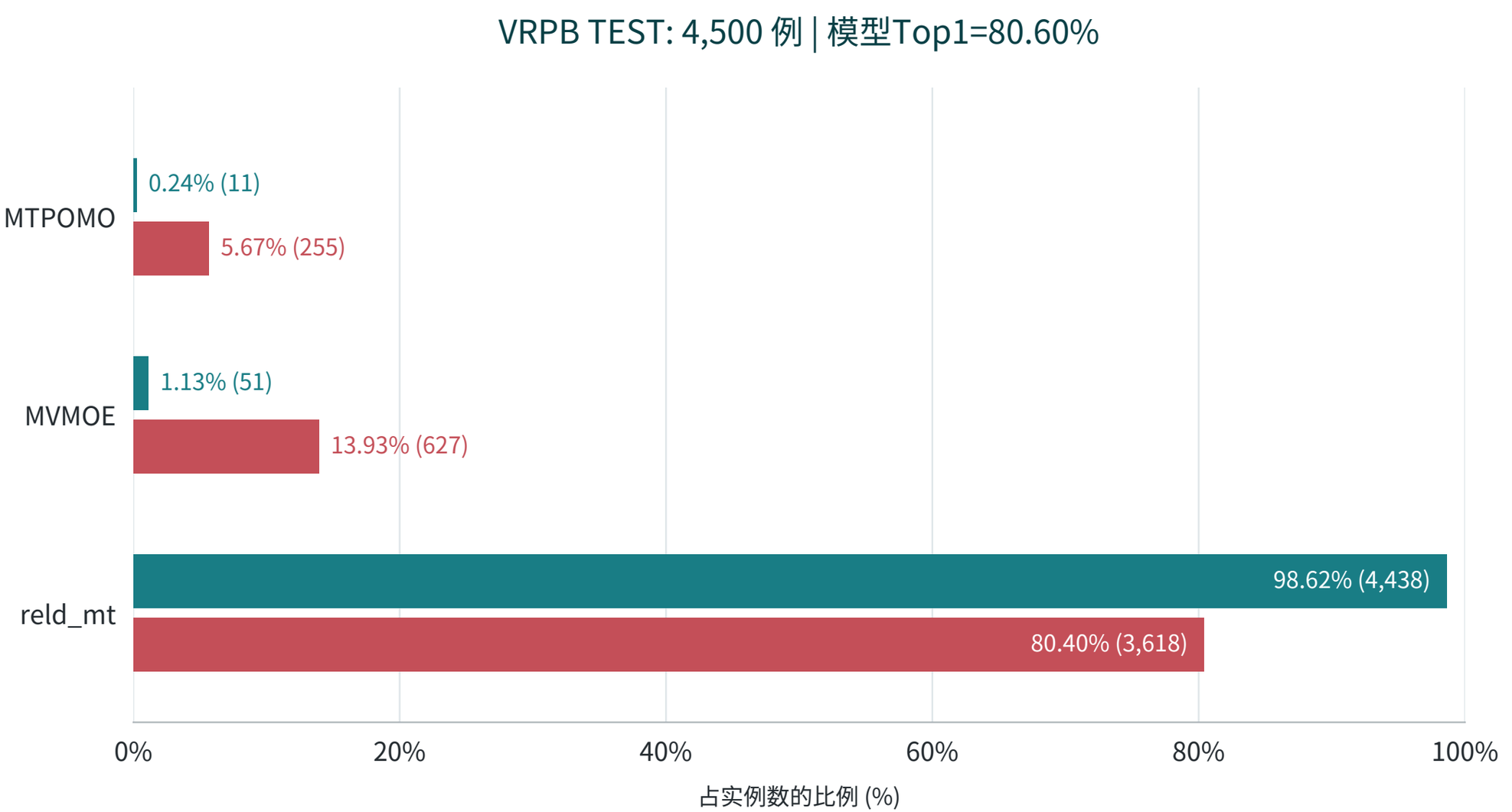
R91/B_utility_probabilities: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/B_utility_probabilities | VRPB | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 cdcb5f1da1e3c30c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

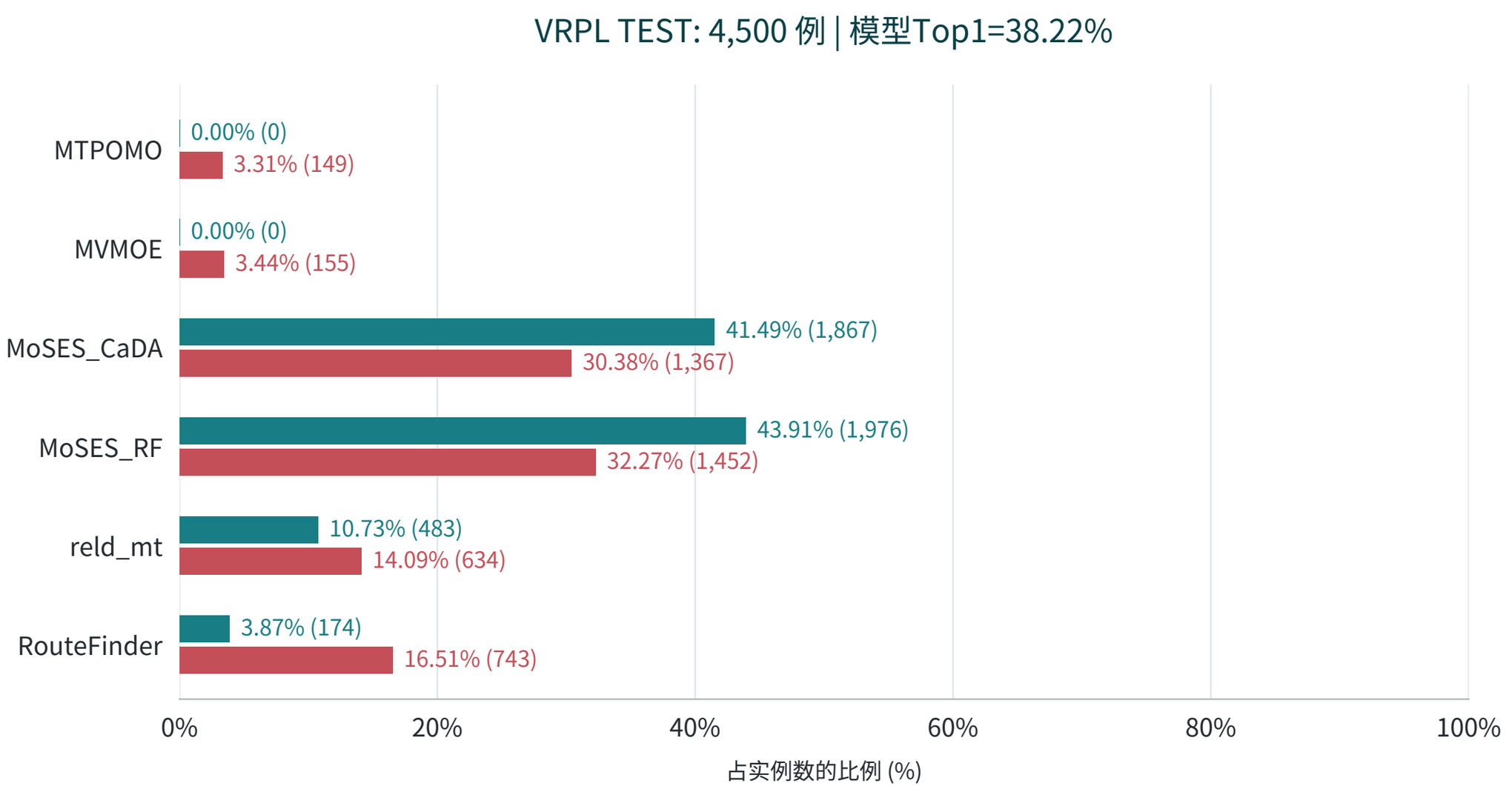
R91/B_utility_probabilities: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/B_utility_probabilities | VRPL | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 cdcb5f1da1e3c30c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

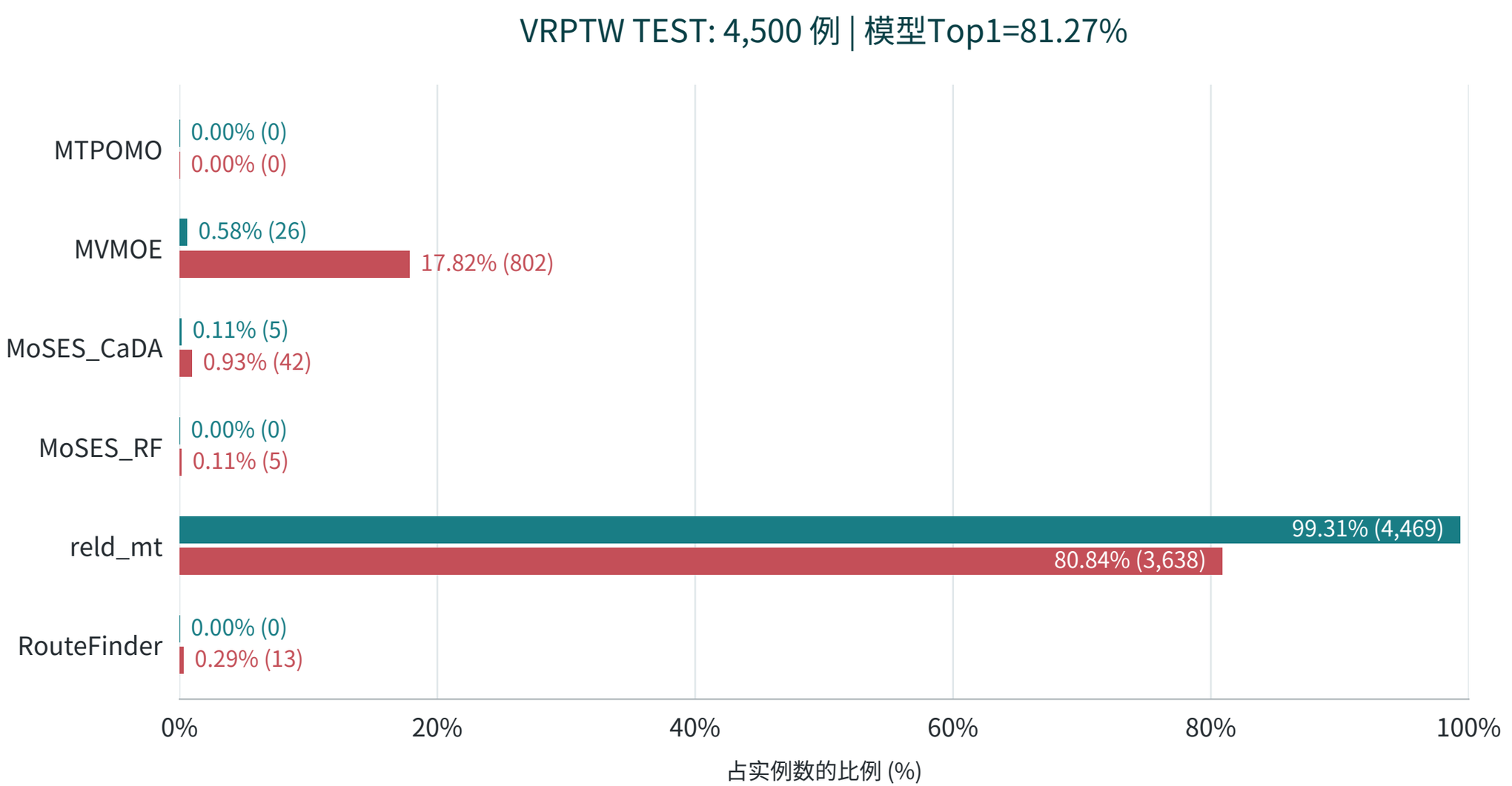
R91/B_utility_probabilities: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/B_utility_probabilities | VRPTW | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 cdcb5f1da1e3c30c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

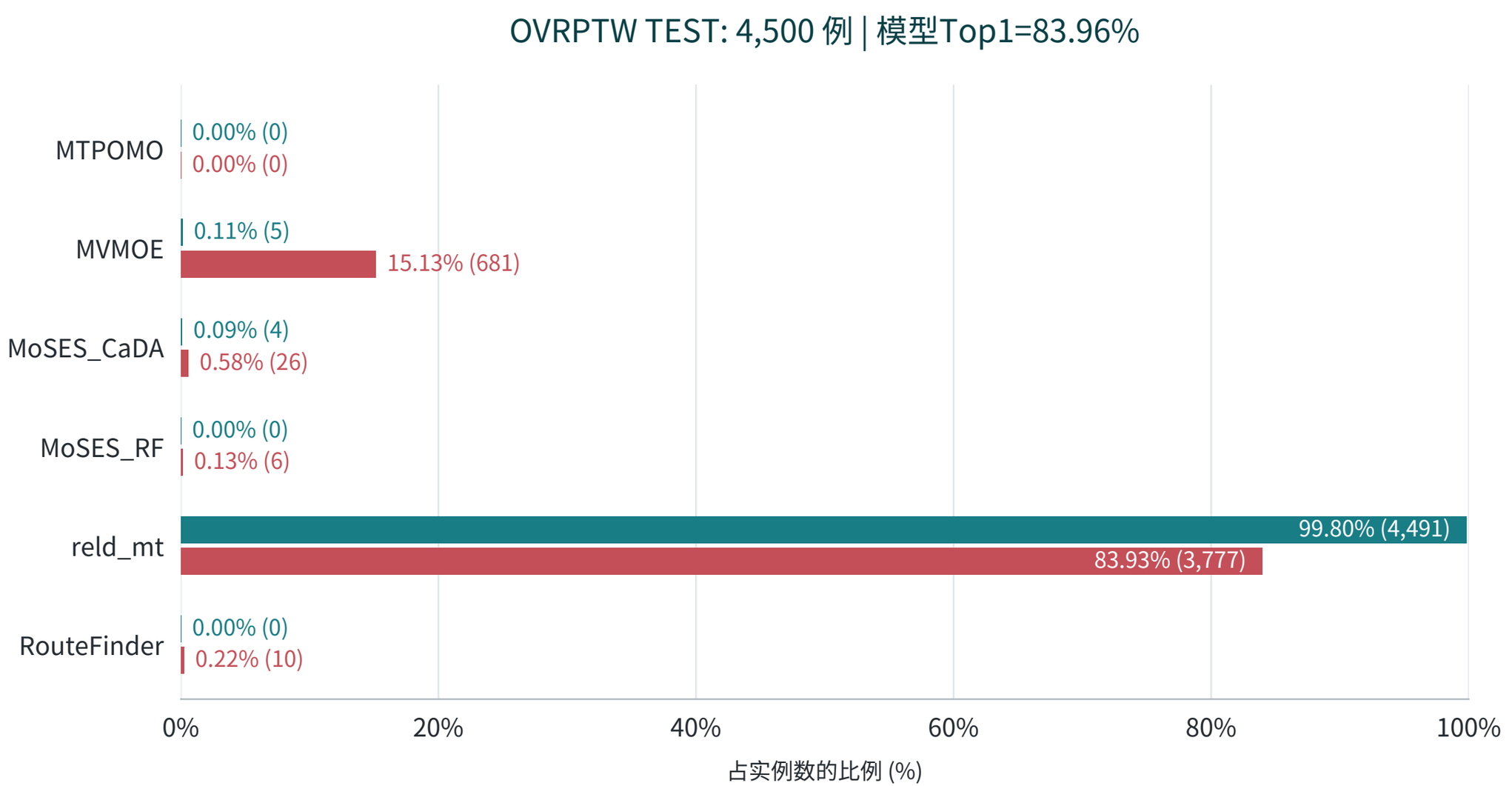
R91/B_utility_probabilities: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/B_utility_probabilities | OVRPTW | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 cdcb5f1da1e3c30c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

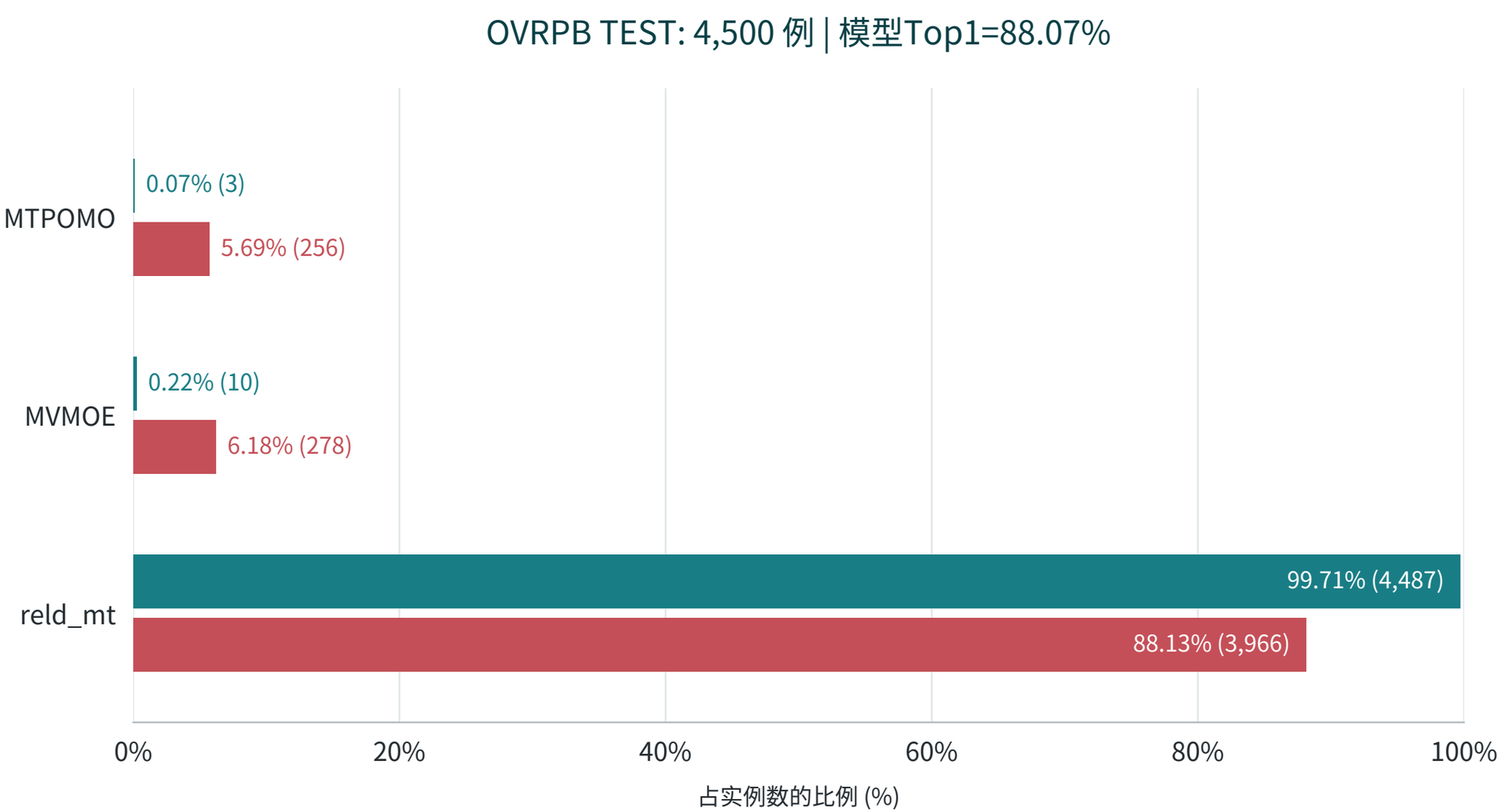
R91/B_utility_probabilities: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/B_utility_probabilities | OVRPB | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 cdcb5f1da1e3c30c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

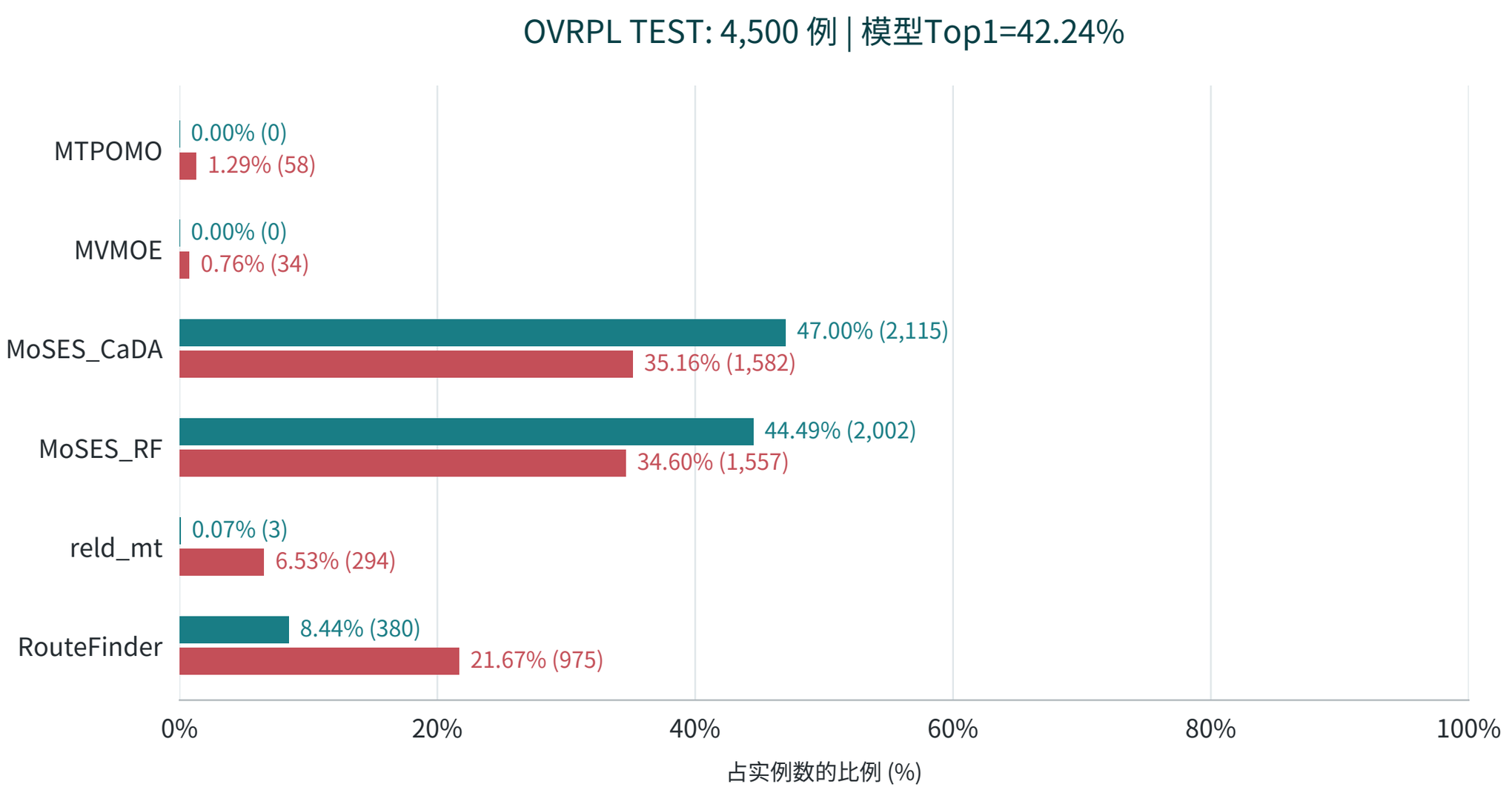
R91/B_utility_probabilities: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/B_utility_probabilities | OVRPL | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 cdcb5f1da1e3c30c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

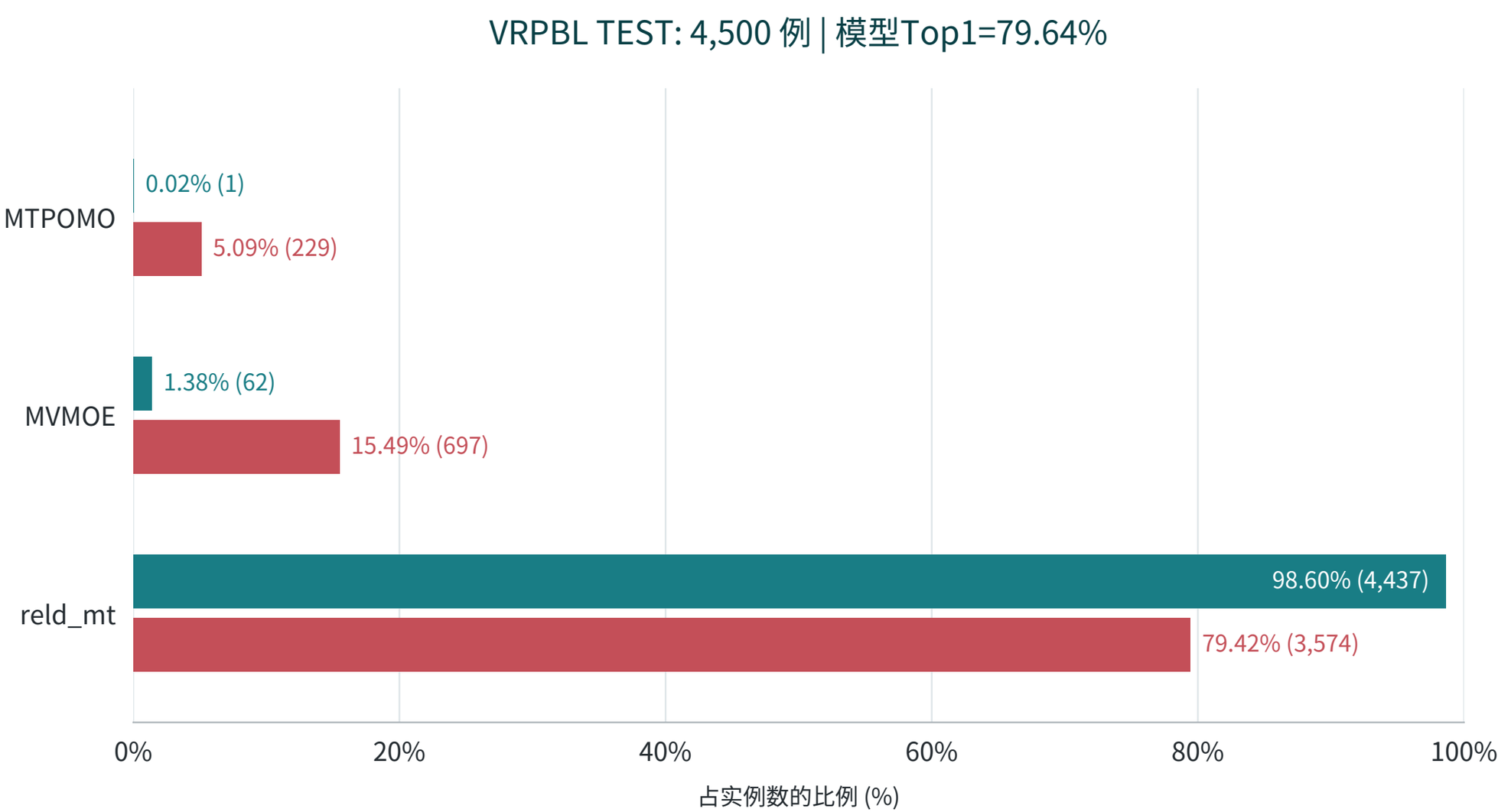
R91/B_utility_probabilities: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/B_utility_probabilities | VRPBL | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 cdcb5f1da1e3c30c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

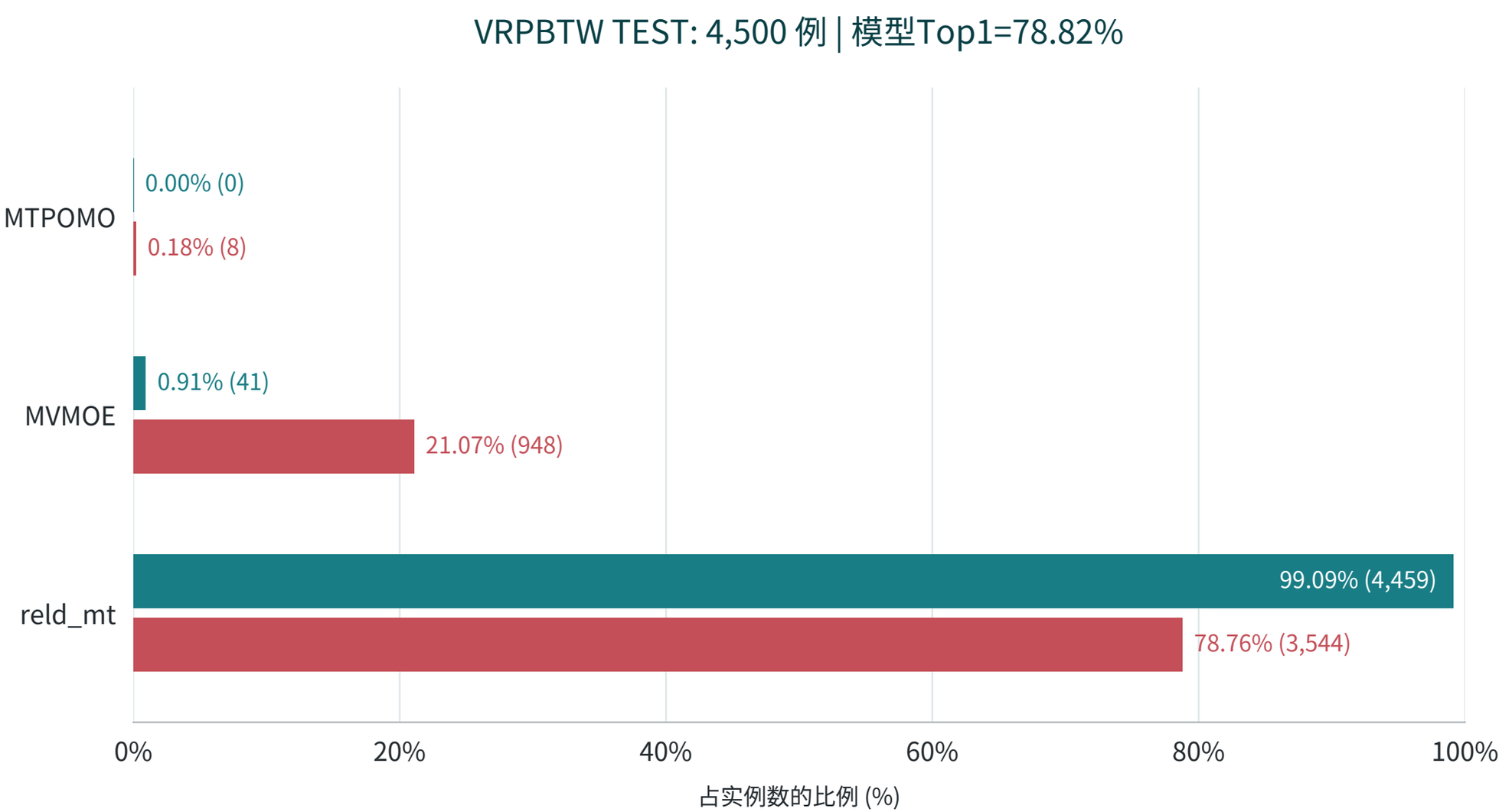
R91/B_utility_probabilities: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/B_utility_probabilities | VRPBTW | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 cdcb5f1da1e3c30c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

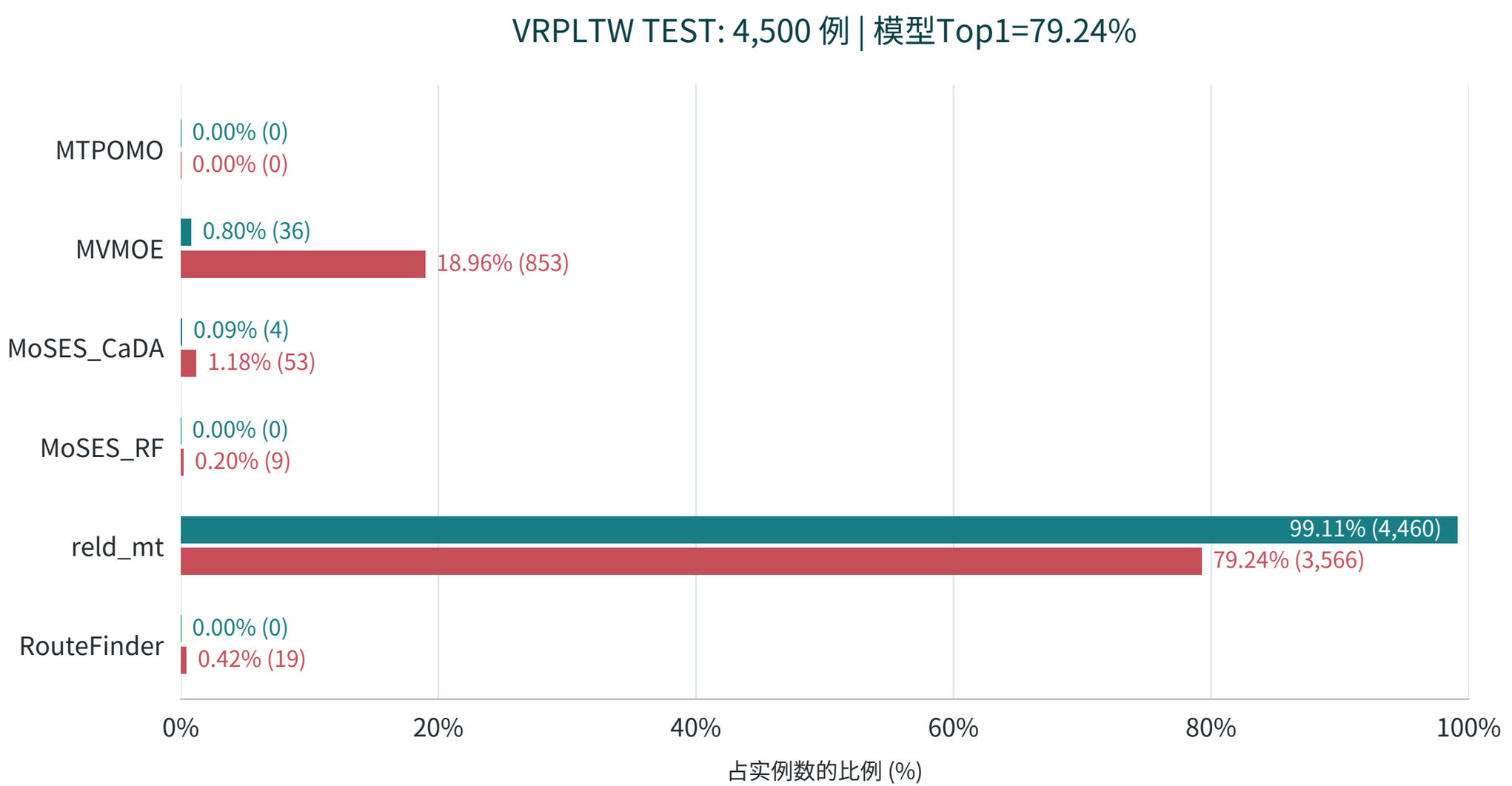
R91/B_utility_probabilities: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/B_utility_probabilities | VRPLTW | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 cdcb5f1da1e3c30c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

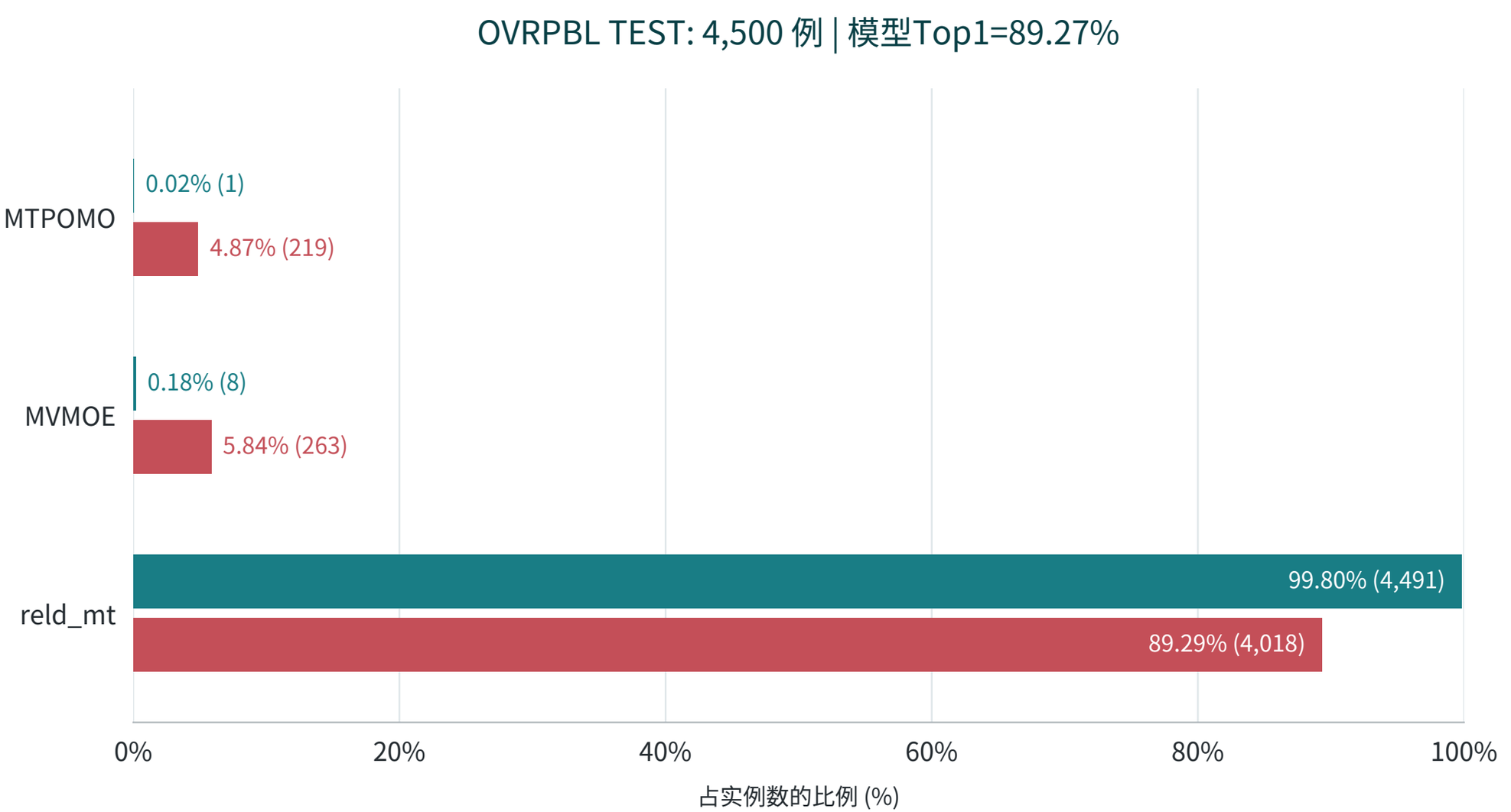
R91/B_utility_probabilities: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/B_utility_probabilities | OVRPBL | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 cdcb5f1da1e3c30c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

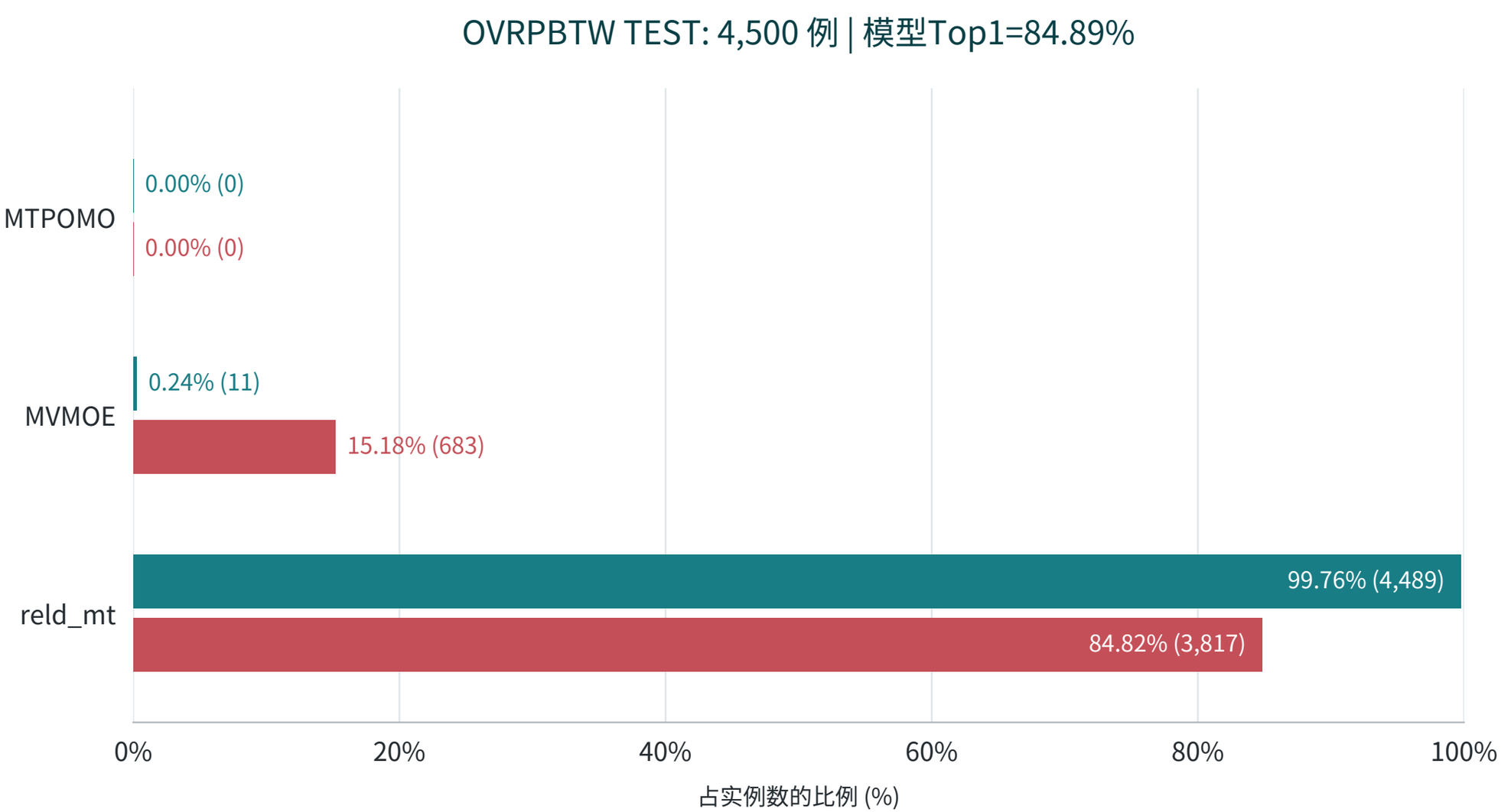
R91/B_utility_probabilities: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/B_utility_probabilities | OVRPBTW | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 cdcb5f1da1e3c30c

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



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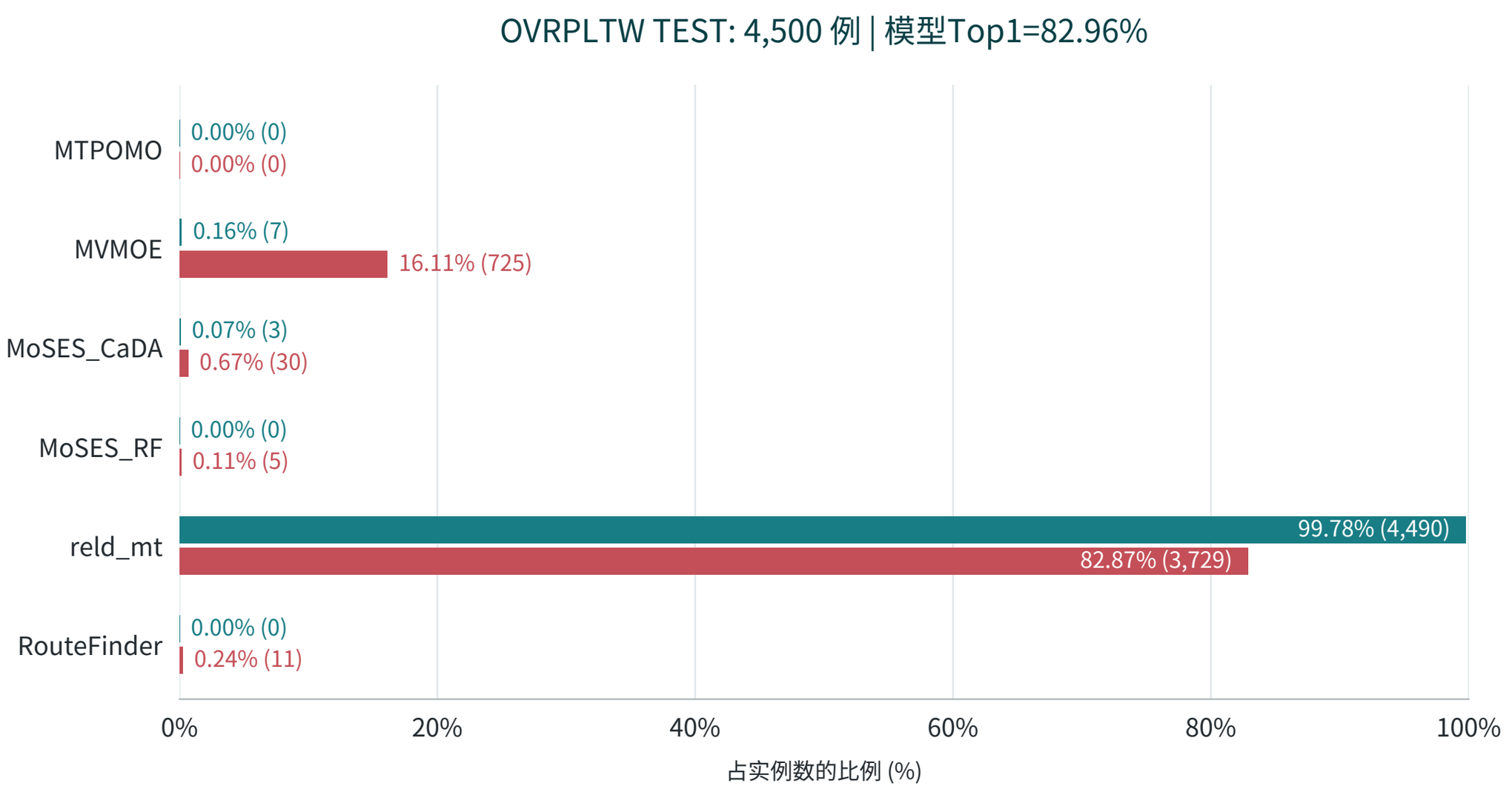
R91/B_utility_probabilities: stable global solver-ID score ties. Same R91 lock; LIB inference only; no training/reselection/Rxx mixing.

R91/B_utility_probabilities | OVRPLTW | 方法分布

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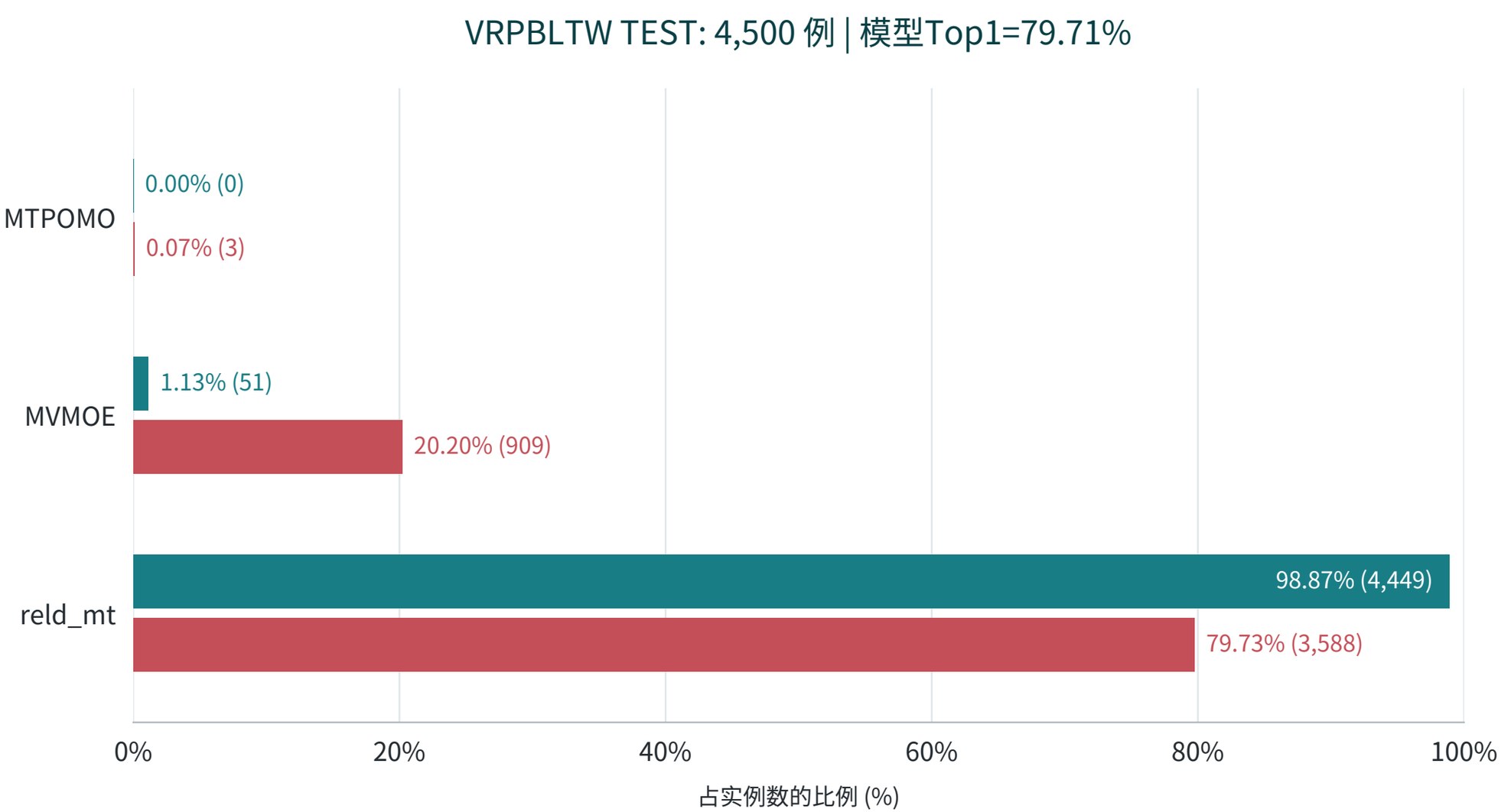
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R91/B_utility_probabilities | VRPBLTW | 方法分布

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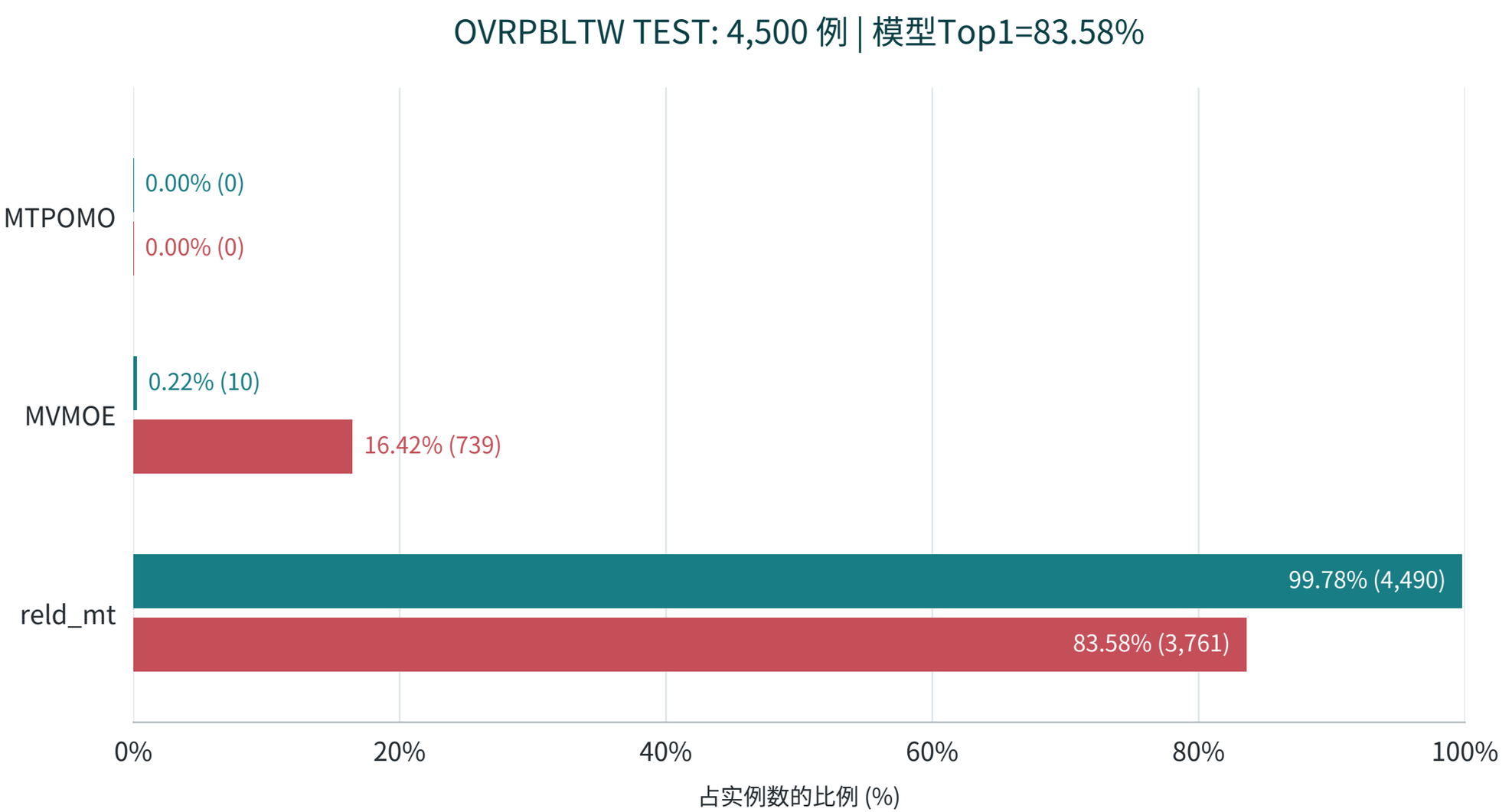
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